

```

1  ///! Data Type Conversion (data type casting) in C++
2  ✓ /*
3      ? convert one data type to another data type is called data type conversion or data type casting.
4
5
6      ~
7          Higher Data Type
8              Long Double
9                  ↓
10                     Double
11                         ↓
12                            Float
13                                ↓
14                                   Long
15                                       ↓
16                                          Int
17                                              ↓
18                                                 Short
19                                                     ↓
20                                                        Char
21
22          ~
23              Lower Data Type
24
25
26
27      # if we convert lower data type to higher data type then it is called implicit type
28      # conversion or automatic type conversion or type promotion.
29      # so (No data loss) in implicit type conversion.

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30
31      ! functions : in (#include <string>) header file we have three functions to convert string to int, float, double.
32
33          ~ 1) stoi(string) : convert string to int
34              # string to int conversion function.
35
36          ~ 2) stof(string) : convert string to float
37              # string to float conversion function.
38
39          ~ 3) stod(string) : convert string to double
40              # string to double conversion function.
41
42          ~ 4) to_string(data) : convert int, float, double to string
43              # int, float, double to string conversion function.
44
45
46      ? convert double to integer :
47
48          double d = 3.14;
49          int i = d; // implicit type conversion (data loss)
50          int j = (int)d; // explicit type conversion (data loss)
51          int k = int (d); // explicit type conversion (data loss)
52          // so i, j, k will be 3 (data loss) same output for all three variables.
53
54      ? convert string to int , float , double :
55
56          string s = "3.14";
57
58          // convert string to integer
59          int i = stoi(s); // implicit type conversion (data loss)
60
61          // convert string to float
62          float f = stof(s); // implicit type conversion (data loss)

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58
59 // convert string to double
60 double d = stod(s); // implicit type conversion (data loss)
61 // so i will be 3 (data loss) and f, d will be 3.14 (no data loss).
62
63
64 ? convert numbers to string :
65     int i = 3;
66     float f = 3.14;
67     double d = 3.14;
68
69 // convert int to string
70 string s1 = to_string(i); // implicit type conversion (no data loss)
71
72 // convert float to string
73 string s2 = to_string(f); // implicit type conversion (no data loss)
74
75 // convert double to string
76 string s3 = to_string(d); // implicit type conversion (no data loss)
77 // so s1 will be "3" and s2, s3 will be "3.140000" (no data loss).
78
79
80 */
81
```